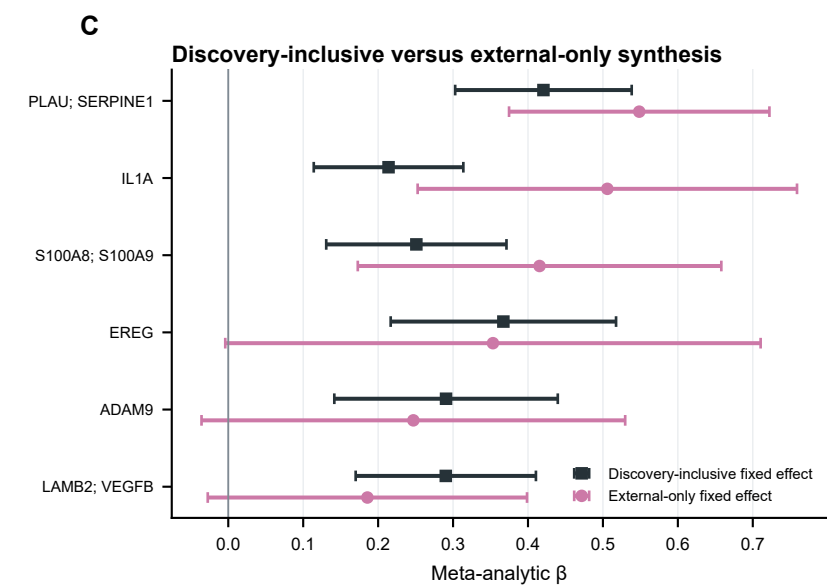
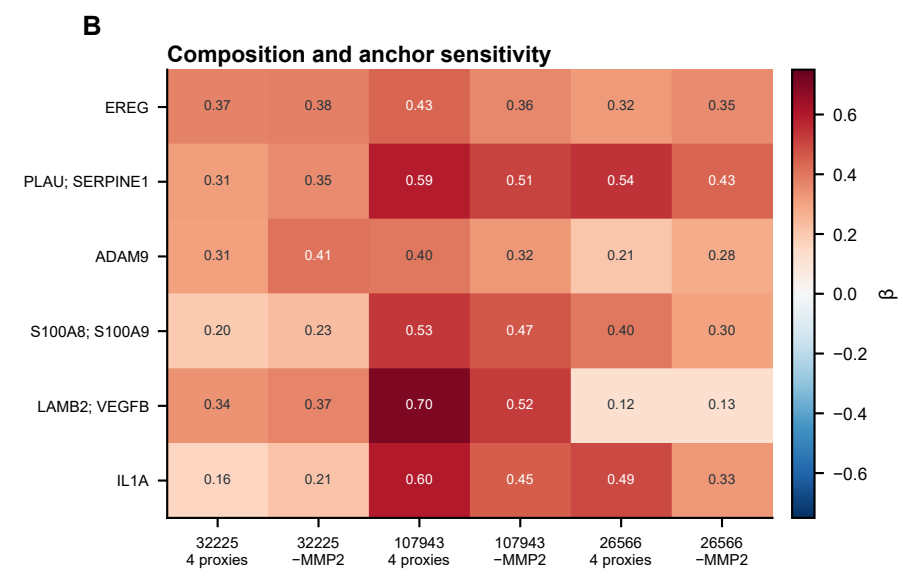
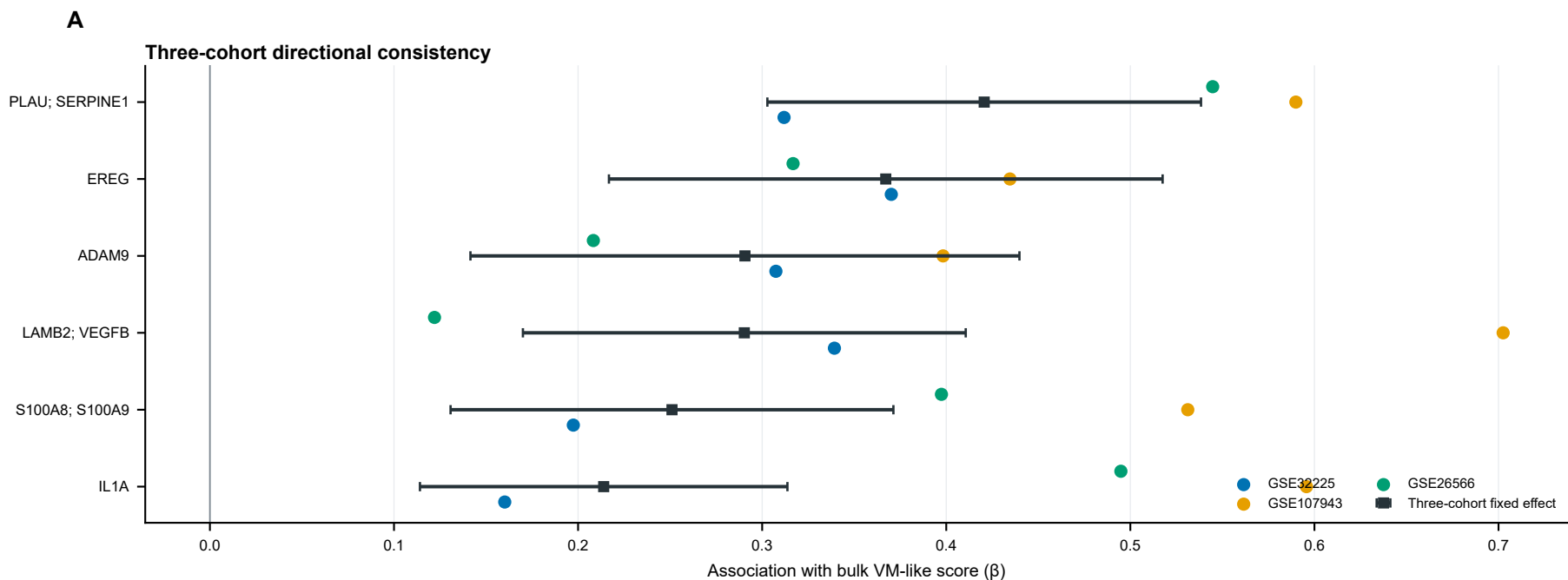


Fig. 4 | Directional bulk consistency and composition-aware sensitivity across iCCA cohorts



All models are patient-sample-level bulk associations adjusted with expression proxies for myeloid, fibroblast and endothelial composition; they do not identify a cellular source or establish causality.